

MSc. Data Management and Analyses Capstone Project Guide

Department of Data Science and Economic Policy

University of Cape Coast

August 11, 2025

Outline

- 1 Introduction & Objectives
- 2 Roles & Responsibilities
- 3 Timeline & Milestones
- 4 Deliverables
- 5 Evaluation & Feedback
- 6 Resources & Support
- 7 Sample Capstone Project Topics

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Introduction & Objectives

The Capstone Project is the culminating experience of the **MSc. Data Management and Analyses** program, designed to integrate theoretical knowledge with practical application.

- Students will work individually to address real-world problems using data-driven approaches or systems.
- The project aims to enhance:
 - Technical expertise
 - Critical thinking
 - Application of knowledge
 - Communication skills

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Roles & Responsibilities

- **Students:** Drive the project, manage timelines, produce deliverables, and present results.
- **Faculty Advisors:** Provide guidance, feedback, and technical support throughout the project lifecycle.
- **External Sponsors (if applicable):** Supply problem statements, data, and domain insights, and review progress.

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Timeline & Milestones

- **August 2025:** Topic selection, proposal draft, project design, and preliminary data exploration.
- **September 1st–24th, 2025:** Model development, analysis, validation, documentation, and final presentation.
- **September 25th–29th, 2025:** Work in progress and presentation of findings.
- **October 15th, 2025:** Submission of final thesis.

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- Hardbound copy of capstone project.

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Evaluation & Feedback

Projects will be assessed on:

- Problem relevance
- Technical depth
- Methodological rigor
- Quality of deliverables
- Clarity of communication

Assessment criteria will focus on innovation, reproducibility, and impact. Feedback will be provided at each milestone.

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- Make time and request samples from the Department's Administrator during your face-to-face lecture periods.
- Consult case studies, research, and experts as needed.

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Sample Topics (1/3)

- 1 Predictive model on digital financial inclusion and sustainability development in Sub-Sahara Africa.
- 2 Building a predictive model on maternity mortality using machine learning: Comparison of different modeling techniques.
- 3 The practice of obtaining patients' consent before surgery: A study of Korle-Bu Teaching Hospital.
- 4 Assessing the disparity in academic performance of pupils in the Basic Education Certificate Examination (BECE) in public and private schools in the Atimatim Circuit, Afigya Kwabre district in the Ashanti Region, Ghana.

Sample Topics (2/3)

- 1 Stakeholder's perspective and key performance indicators of Ghana's Free Senior High School Policy – Case study of Fumbisi Senior High Agric School.
- 2 Ghana Revenue Authority's Integrated Customs Management Systems (ICMS): Prospects, challenges and recommendations.
- 3 Using chatbot to provide better customer service for UNICOF.
- 4 Unraveling the influence of compensation, benefits, gender and years of service on employee performance and job satisfaction: A comprehensive study.

Sample Topics (3/3)

- 1 Analysing customer churn with machine learning.
- 2 Patients' satisfaction level and challenges in accessing health care at Asankrangwa Catholic Hospital.
- 3 The practice of obtaining patients' consent before surgery: A case study at the Korle-Bu Teaching Hospital.
- 4 Determinants of demand for homeowners insurance policy in the Ho municipality in Ghana.